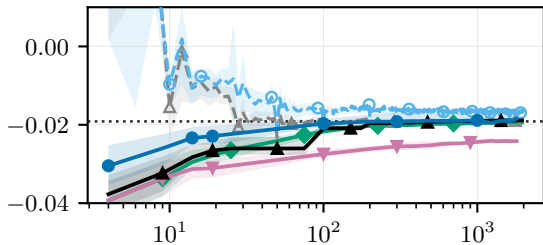
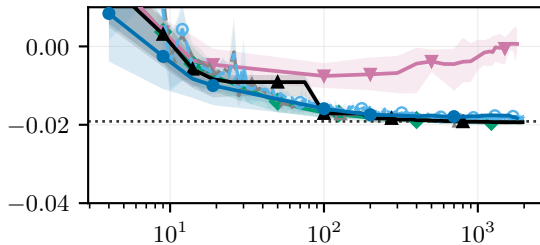
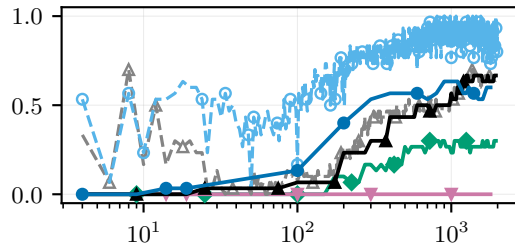


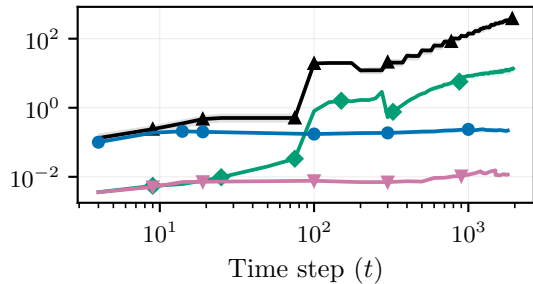
In-sample objective value



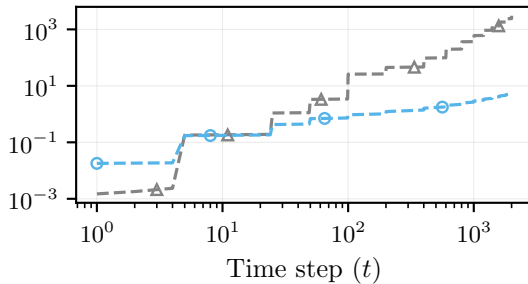
Out-of-sample objective value

Confidence $1 - \hat{\beta}_t$ 

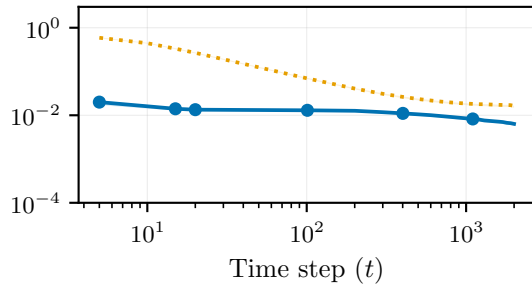
Per-iteration time (s), exact



Cumulative time (s), online



Online clustering dynamic regret



● online clustering

▲ DRO

◆ SAA

..... H^*

..... upper bound

○- online clustering subgrad

-▲- DRO subgrad

▼ cluster SAA

● empirical regret